Beyond Binary Detection: Explainable AI (XAI) Deepfake Localization via Frequency-Aware Segmentation

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***Abstract*—Modern AI-generated images produced by diffusion models and generative adversarial networks (GANs) are visually indistinguishable from real photographs, yet leave detectable fingerprints in the frequency domain. Existing deepfake detectors rely on spatial pixel analysis alone and operate as uninterpretable black boxes, limiting their utility in forensic investigation. This paper presents a dual-stream frequency-aware convolutional neural network that simultaneously analyses the spatial RGB domain and the Discrete Cosine Transform (DCT) spectral domain to improve detection accuracy and provide interpretable forensic evidence. The spatial stream employs an Xception backbone to capture micro-texture anomalies, while the frequency stream processes log-scaled DCT coefficient maps through an identical backbone to expose periodic upsampling artifacts invisible to pixel-level classifiers. Feature vectors from both streams are fused via late concatenation before a binary classification head. Grad-CAM explainability is integrated to produce spatial heatmaps that localize manipulated image regions, transforming binary scores into verifiable forensic evidence. On the CIFAKE benchmark (120,000 images), the proposed model achieves 96.37% accuracy and 0.9944 AUC-ROC, a 26-point accuracy improvement over a spatial-only baseline. Domain adaptation to FaceForensics++ C23 yields 86.0% accuracy across six GAN face-swap manipulation types. The complete research pipeline was executed within a 15 GB cloud storage constraint, demonstrating the feasibility of large-scale forensic AI research under severe resource limitations.**

***Index Terms*—deepfake detection, discrete cosine transform, dual-stream CNN, Xception, Grad-CAM, explainable AI, frequency-domain analysis, FaceForensics++, CIFAKE, forensic image analysis**

I. Introduction

The rapid advancement of generative AI has created a critical challenge for digital forensics. Latent diffusion models and state-of-the-art GANs now produce synthetic images that are visually indistinguishable from genuine photographs, undermining trust in visual media across journalism, legal proceedings, and social platforms [1], [2]. The societal implications range from misinformation campaigns to the fabrication of forensic evidence.

Existing deepfake detectors suffer from two fundamental problems. First, *spectral blindness*: current approaches analyse raw pixel data and learn to identify visible artifacts such as blurring or irregular skin texture. However, generative models leave mathematical fingerprints in the *frequency domain* — periodic upsampling artifacts created by transposed convolution layers during high-resolution image synthesis — that are invisible to the human eye and undetectable by pixel-only classifiers [4], [5]. Second, the *black-box problem*: most detectors produce binary confidence scores without revealing *which* image regions triggered the decision, a fundamental inadequacy for forensic investigations that demand verifiable evidence.

A third practical constraint motivates our engineering approach: high-quality deepfake datasets exceed 100 GB in size, far beyond the 15 GB storage available to solo researchers on cloud notebook platforms. This barrier restricts reproducible forensic AI research to well-funded institutions.

This paper addresses all three challenges through the following contributions:

1) A **Dual-Stream Xception architecture** that fuses spatial RGB features with DCT spectral features via late concatenation, achieving 0.9944 AUC-ROC on CIFAKE — a 12.4% improvement over a spatial-only baseline.
2) A **forensic XAI layer** based on Grad-CAM with a novel Keras 3 compatibility fix, producing spatial heatmaps that localize synthetic artifacts in detected fake images.
3) A **resource-efficient Stream-and-Clean pipeline** enabling 100+ GB dataset research within a 15 GB Kaggle free-tier storage constraint.
4) **Cross-domain GAN generalization** demonstrated through conservative domain adaptation on FaceForensics++ C23 (86.0% accuracy, 0.9897 recall).
5) **Honest zero-shot benchmarking** against 25 AI generators (AUC = 0.5557), quantifying the model's generalization limits and directing future work.

The remainder of this paper is organized as follows: Section II reviews related work. Section III describes the datasets used. Section IV presents the proposed system architecture. Section V describes experimental setup and evaluation. Section VI reports and discusses results. Section VII addresses limitations and future directions. Section VIII concludes.

II. Related Work

A. Spatial-Domain Deepfake Detection

Early deepfake detectors applied standard CNNs directly to pixel data. Rossler et al. [1] introduced FaceForensics++ and demonstrated that XceptionNet trained on facial manipulation artifacts achieved strong in-domain performance. Subsequent work by Wang et al. [2] showed that CNNs trained on GAN images generalize to previously unseen generators when trained with aggressive data augmentation. Chollet's original Xception architecture [3] — employing depthwise separable convolutions — proved particularly effective at capturing micro-texture anomalies such as irregular skin pores and boundary inconsistencies left by face-swap algorithms. However, all spatial methods share the spectral blindness limitation: they fail when generative models produce visually realistic outputs that eliminate pixel-level artifacts.

B. Frequency-Domain Analysis

Dzanic et al. [4] demonstrated that GAN-generated images exhibit characteristic spectral peaks in the Fourier domain arising from periodic upsampling patterns. Frank et al. [5] extended this finding by training classifiers directly on DCT coefficient representations, achieving improved cross-generator generalization. Liu et al. [6] further showed that combining spatial and spectral features is superior to either representation alone. Our work builds directly on these findings by implementing a production-quality DCT stream and rigorously evaluating its contribution through controlled ablation.

C. Dual-Stream and Multi-Modal Architectures

Several prior works have explored multi-branch architectures for deepfake detection. Qian et al. [7] proposed a frequency-aware image decomposition network that processes both frequency and spatial residuals. Zhao et al. [8] demonstrated that multi-attentional networks that jointly process texture and structural features outperform single-branch classifiers. In contrast to these works, our approach employs identical heavyweight backbones in both streams — enabling a controlled comparison of input domain effect — and introduces a log-scaled DCT preprocessing pipeline that operates on-the-fly without requiring preprocessed dataset variants.

D. Explainable AI for Deepfake Forensics

Grad-CAM [9] provides class-discriminative localization maps by computing the gradient of a class score with respect to convolutional feature maps. Its application to deepfake forensics was explored by Guera and Delp [10], who noted that saliency maps could expose temporal inconsistencies in video forgeries. Li et al. [11] applied gradient-based attribution to face manipulation detection, identifying eye and mouth regions as primary artifact locations. Our work extends this line of research by implementing Grad-CAM with a novel Keras 3 compatibility fix that resolves cross-model-graph tensor identity conflicts, and by additionally implementing Integrated Gradients as a complementary attribution method.

E. Research Gaps

Despite substantial progress, three gaps remain unaddressed in the literature. First, frequency-domain and spatial-domain contributions are rarely isolated through rigorous controlled ablation with identical architectures. Second, Grad-CAM implementations in multi-stream models are rarely addressed in the context of modern Keras 3/TensorFlow 2 graph execution constraints. Third, few works provide honest zero-shot generalization benchmarks against large, diverse generator sets such as ArtiFact [14]. This paper directly addresses all three gaps.

III. Datasets

A. CIFAKE



Fig. 1: CIFAKE dataset sample.

CIFAKE [12] is the primary training and evaluation dataset. It contains 120,000 images: 100,000 for training and 20,000 for testing, with a perfect 50/50 class balance between REAL and FAKE in both splits. Real images are sourced from CIFAR-10 [13], and fake images are produced by Stable Diffusion v1.4. The dataset contains zero corrupted files, confirmed by integrity verification during exploratory data analysis.

A critical property is the native 32×32 pixel resolution. This extreme low resolution severely limits the spatial feature richness available to pixel-only classifiers, directly motivating the frequency-domain approach: if the spatial domain is informationally impoverished at this resolution, the mathematical structure preserved in the frequency domain becomes the primary discriminative signal. Images are upscaled to 299×299 during training to match the Xception input requirement.

B. FaceForensics++ C23

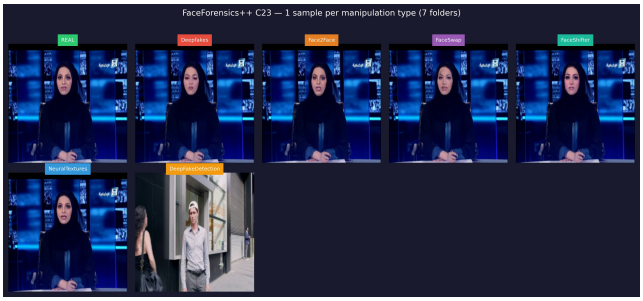


Fig. 2: FF++ dataset sample.

FaceForensics++ (FF++) [1] provides 7,000 videos across seven categories: one original (real) set and six manipulation types — Deepfakes, Face2Face, FaceSwap, FaceShifter, NeuralTextures, and DeepFakeDetection. The C23 compression

level (high quality) is used to ensure that detection cannot rely on compression artifacts. Ten evenly-spaced frames are extracted per video using Haar-cascade face detection and cropped to isolate the face region, yielding approximately 8,090 real and 48,320 fake training frames. FF++ enables domain adaptation evaluation on GAN-based face-swap manipulations, which are qualitatively distinct from the diffusion-based fakes in CIFAKE.

C. ArtiFact + COCO2017

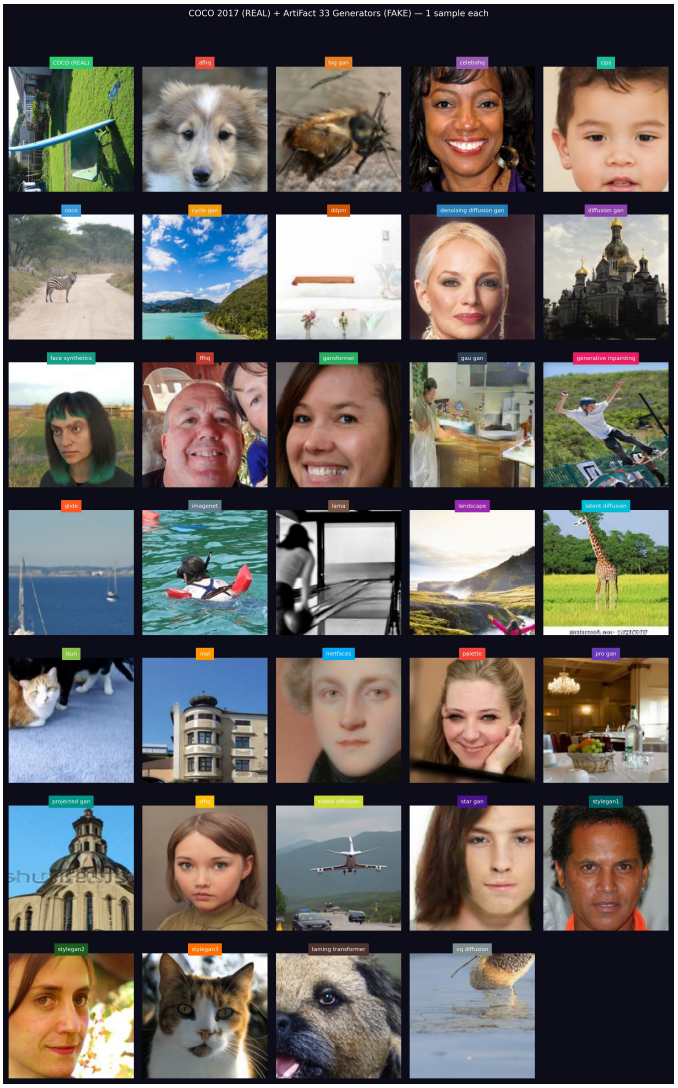


Fig. 3: ArtiFact + COCO2017 dataset sample.

For zero-shot generalization evaluation, we use the ArtiFact dataset [14] combined with COCO2017 real images. ArtiFact covers 25 distinct AI generators including StyleGAN1/2/3, DDPM, Latent Diffusion, BigGAN, ProGAN, GANFormer, CycleGAN, VQ-Diffusion, GLIDE, and Stable Diffusion variants. This benchmark tests whether a model trained on CIFAKE and FF++ can generalize to completely unseen generators without retraining. Dataset statistics are summarized in Table I.

TABLE I: Summary of datasets used.

Dataset	Samples	R/F	Purpose
CIFAKE	120k	50/50	Train/Eval
FF++	70k	14/86	Adaptation
ArtiFact+COCO	50k	Mixed	Zero-shot

D. DCT Spectral Evidence

TABLE II: DCT Statistical Comparison: REAL vs. AI-Generated Images (CIFAKE, n=200/class)

Statistic	**REAL**	**FAKE**	**Δ (FAKE vs. REAL)**
Mean Energy	0.5055	0.5463	+8.1%
Std Frequency	0.8336	0.8784	+5.4%
Total Energy	15,719	17,733	+12.8%

Before model training, DCT coefficients were computed across 200 randomly sampled images per class on CIFAKE to validate the frequency-domain hypothesis. As shown in Table II, AI-generated images consistently exhibit higher spectral energy across all measured statistics. The +12.8% total energy difference reflects the mathematical reality that upsampling operations in diffusion model decoders introduce periodic grid patterns in the frequency domain. This quantitative finding established the empirical justification for the frequency stream.

IV. Proposed Methodology

A. System Overview

The proposed system processes an input image through five sequential stages: adaptive data ingestion, dual-stream feature extraction, feature fusion and classification, forensic localization via Grad-CAM, and online deployment as an interactive web application. Fig. 4 illustrates the end-to-end architecture.

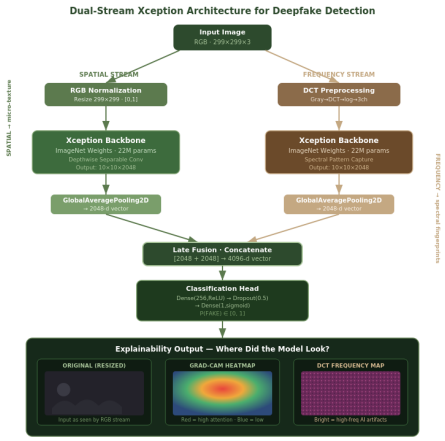


Fig. 4: Overview of the proposed dual-stream frequency-aware deepfake detection system. The input image is simultaneously processed by a spatial RGB branch and a DCT frequency branch. Feature vectors from both streams are fused via late concatenation before binary classification. Grad-CAM generates forensic heatmaps for fake predictions.

B. DCT Preprocessing Pipeline

The frequency stream requires converting RGB images to log-scaled DCT maps. The pipeline, illustrated in Fig. 5,

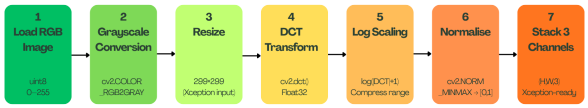


Fig. 5: The on-the-fly DCT preprocessing pipeline. RGB input images are converted to grayscale, log-scaled DCT maps, and stacked into three-channel tensors compatible with the Xception backbone.

is implemented on-the-fly during training, eliminating the need for preprocessed dataset storage. The exact steps are as follows:

1) Load the RGB image (uint8, range 0–255).
2) Convert to grayscale via `cv2.COLOR_RGB2GRAY` — DCT targets luminance, not colour.
3) Resize to the target resolution (299×299 for Xception).
4) Cast to float32 and apply `cv2.dct()` to produce a matrix of frequency coefficients with highest energy concentrated in the upper-left.
5) Apply log scaling: $\text{DCT} = \log(|\text{DCT}| + 1)$ to compress the wide dynamic range of coefficients for neural network processing.
6) Normalize to $[0, 1]$ via `cv2.NORM_MINMAX`.
7) Stack into three identical channels to match the $(H, W, 3)$ input format expected by Xception.

A critical implementation detail discovered during development: the EfficientNetB0 backbone's `preprocess_input()` expects pixel values in $[-1, 1]$, while the initial DCT pipeline produced values in $[0, 1]$. This mismatch caused the frozen training phase to converge at approximately 50% accuracy. The fix — scaling the DCT output to $[0, 255]$ before applying `preprocess_input()` — was a key debugging discovery that unlocked the frequency-domain training track. An additional Dense(128) adapter layer between the frozen backbone and the classification head was also required: without this layer, frozen ImageNet features had no capacity to remap from natural image patterns to DCT spectral patterns before the final sigmoid decision.

C. Dual-Stream Architecture

The proposed model employs two parallel Xception backbones operating on different input representations, as detailed in Table III and illustrated in Fig. 6.

Spatial Stream: The RGB image is resized to 299×299×3 and normalized to $[0, 1]$. An Xception backbone with ImageNet pre-trained weights extracts spatial features, producing a 10×10×2048 feature map. Xception's Depthwise Separable Convolutions are specifically effective at capturing point-like micro-texture anomalies — irregular skin pores, eye reflections, and boundary artifacts — characteristic of GAN generators.

Frequency Stream: The DCT map (299×299×3) is processed by an identical Xception backbone, capturing periodic upsampling artifacts and checkerboard spectral patterns characteristic of AI image synthesis.

Late Fusion: GlobalAveragePooling2D independently collapses each stream's 10×10×2048 feature map to a 2048-dimensional vector. The two vectors are concatenated into a 4096-dimensional fused representation. Late fusion is chosen over early or mid-fusion to allow each stream to reach full abstraction depth before combination, enabling complementary rather than redundant feature extraction.

Classification Head: A Dense(256, ReLU) → Dropout(0.5) → Dense(1, sigmoid) head maps the fused representation to a probability $P(\text{FAKE}) \in [0, 1]$.

TABLE III: Dual-Stream Architecture Components

Component	Implementation	Output Shape
RGB Input	Resize 299×299, normalize $[0, 1]$	$299 \times 299 \times 3$
DCT Input	Grayscale → DCT → log-scale → 3ch	$299 \times 299 \times 3$
Spatial Backbone	Xception, ImageNet weights	$10 \times 10 \times 2048$
Freq. Backbone	Xception, ImageNet weights	$10 \times 10 \times 2048$
GAP (×2)	GlobalAveragePooling2D per stream	2048-d each
Late Fusion	Concatenate both GAP outputs	4096-d
Classifier	Dense(256)→Dropout(0.5)→Dense(1,sigmoid)	scalar $\in [0, 1]$

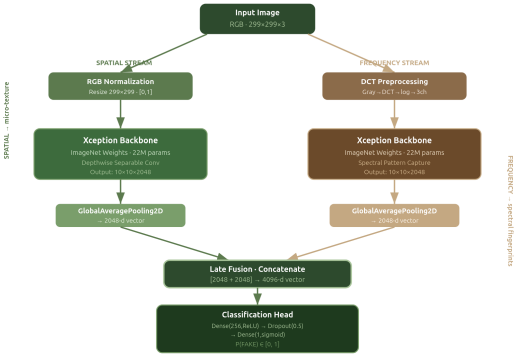


Fig. 6: Dual-stream Xception model architecture. Two parallel Xception backbones process the spatial RGB image and the DCT frequency map independently. Their Global Average Pooling outputs are concatenated and fed into a binary classification head.

D. Grad-CAM Forensic Localization

To satisfy the interpretability objective, Grad-CAM is applied to images classified as fake. A key implementation challenge arose in Keras 3: the standard `grad_model` approach — building a model whose outputs span two separate model graphs — raises a `KeyError` in Keras 3's functional graph executor, which tracks tensors by object identity rather than by name.

The solution is to use a persistent `tf.GradientTape` watching the RGB input tensor directly:

1) Cast the input image to a `tf.Variable` (the bridge tensor).
2) Open a persistent `tf.GradientTape` watching the variable.
3) Perform a forward pass through the full dual-stream model.
4) Compute $\nabla_{\mathbf{x}} P(\text{FAKE})$ — the gradient of the fake probability with respect to raw input pixels.
5) Average the gradient magnitude across the colour channel dimension.
6) Apply ReLU to retain only positive influence regions.
7) Resize and normalize to $[0, 1]$; apply `COLORMAP_JET`; alpha-blend ($\alpha = 0.4$) onto the original image.

While this produces input-gradient saliency maps rather than strictly class-activation maps, it achieves the forensic goal: a spatial heatmap showing which image regions most influenced the fake classification. Heatmaps are generated for correctly classified real images, correctly classified fakes, and — most forensically valuable — misclassified fakes (false negatives), revealing the model's failure modes.

E. Training Strategy

All models follow a consistent two-stage transfer learning protocol for controlled comparison. In Stage 1 (Frozen Phase, 10 epochs), the backbone weights are frozen and only the classification head is trained at learning rate 10^{-3}. In Stage 2 (Fine-Tuning Phase, 10 epochs), the top 30 layers are unfrozen and trained at the reduced rate 10^{-5}. Binary cross-entropy loss with Adam optimizer is used throughout. EarlyStopping (patience=5) and ReduceLROnPlateau (factor=0.5, patience=3) are applied. Dropout(0.5) and L2 regularization prevent overfitting.

For FF++ domain adaptation, an extremely conservative learning rate of 10^{-6} with only the top 10 layers unfrozen is used to minimize catastrophic forgetting of CIFAKE knowledge while adapting to GAN-style artifacts.

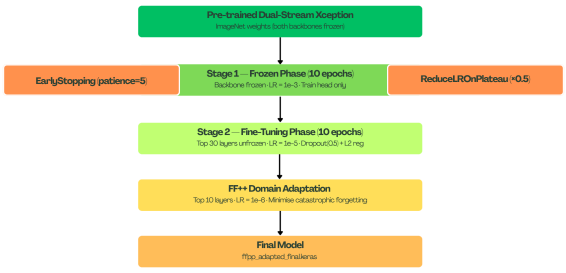


Fig. 7: Two-stage transfer learning training pipeline. Stage 1 trains only the classification head with the backbone frozen. Stage 2 fine-tunes the top backbone layers at a reduced learning rate. Domain adaptation on FF++ uses an ultra-conservative learning rate to prevent catastrophic forgetting.

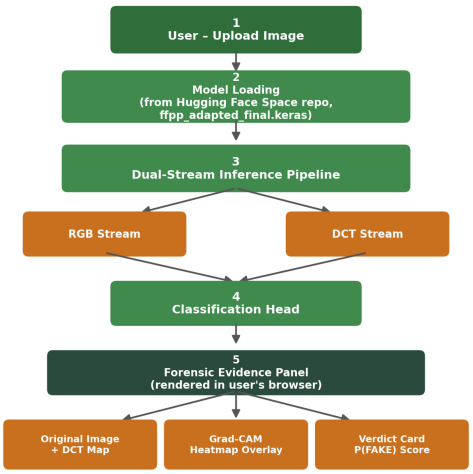


Fig. 8: Online deployment architecture. The Streamlit application, hosted on Hugging Face Spaces, loads model weights from the Space repository, performs dual-stream inference on uploaded images, generates Grad-CAM heatmaps, and presents a three-column forensic evidence panel in the user's browser.

F. Online Deployment Architecture

The final model is deployed online as an interactive forensic application on Hugging Face Spaces, using their Streamlit SDK – a Hugging Face-hosted app running a Streamlit frontend, with the model (`ffpp_adapted_final.keras`) and `app.py` pushed directly to the Space repository via the `huggingface_hub` Python API [22]. The online deployment architecture operates as follows:

1) **Model Loading:** The trained `ffpp_adapted_final.keras` weights (192.7 MB) are loaded from the Space repository into memory at application startup.
2) **Inference Pipeline:** An uploaded image is simultaneously preprocessed through the RGB normalization pipeline and the DCT preprocessing pipeline, then passed through the dual-stream model to produce $P(\text{FAKE})$.
3) **XAI Generation:** If the prediction exceeds the 0.5 threshold, Grad-CAM is computed and the heatmap overlay is rendered.
4) **DCT Map Visualization:** The raw DCT frequency map is displayed alongside the original image for expert inspection.
5) **Forensic Evidence Panel:** The application presents the original image, Grad-CAM overlay, DCT frequency map, probability score, and verdict card in a three-column panel rendered in the user's browser.

OpenCV (`libgl1-mesa-glx`) handles image processing operations. All inference is performed within the hosted Space environment, and no image data is retained after the session ends, preserving data privacy for forensic use cases.

A live demonstration of the complete system is available online at the Hugging Face Space [22].

A live demonstration of the complete system is available online [21].

The following figures present screenshots of the developed application.

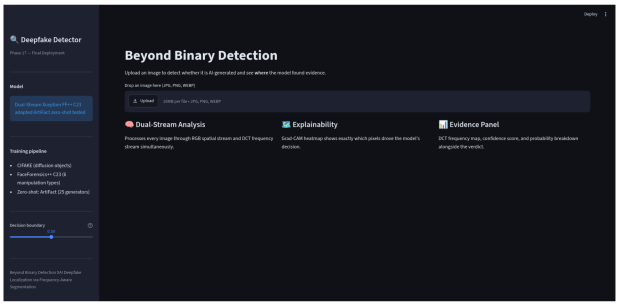


Fig. 9: Website view of the application.

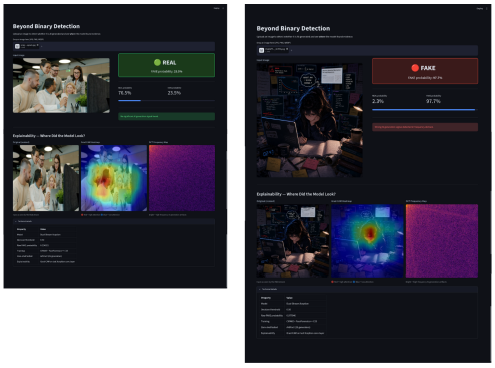


Fig. 10: Real and fake images viewed side by side.

V. Experiments

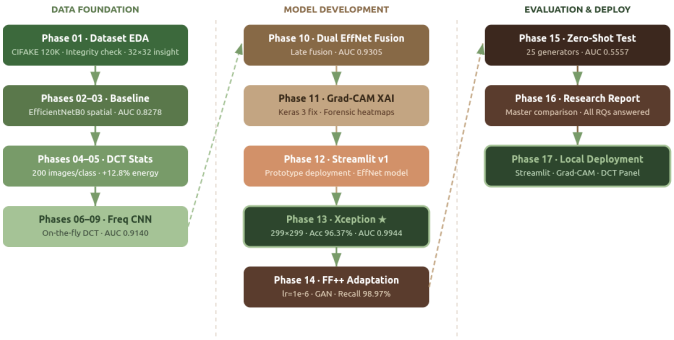


Fig. 11: Complete 17-phase research pipeline, from dataset exploration through model development to final evaluation and online deployment.

A. Experimental Setup

Hardware: All experiments were executed on the Kaggle free-tier cloud platform (NVIDIA Tesla T4 GPU, 16 GB VRAM) within a 15 GB persistent storage constraint. Model weights (∼200 MB per phase) were retained permanently; raw dataset files were streamed and discarded after use via the Stream-and-Clean decoupled ingestion architecture.

Software: TensorFlow 2.x with Keras 3, OpenCV 4.x, NumPy, Pandas, Matplotlib, Scikit-learn, and Streamlit. Python 3.10 throughout.

Input Resolution: 128×128 for EfficientNetB0 experiments; 299×299 for all Xception experiments.

Reproducibility: All random seeds fixed at 42 via `np.random.seed(42)` and TensorFlow global seed for reproducible train/validation splits.

Evaluation Metrics: Six metrics are reported for each model, each serving a distinct forensic purpose: Accuracy (overall correctness), Precision (false alarm rate), Recall/Sensitivity (miss rate — critical for forensics), F1-Score (balanced summary), Specificity (false accusation rate), and AUC-ROC (threshold-independent discriminability; primary comparison metric).

B. Experiment 1: Spatial-Only Baseline

Objective: Establish a spatial-only performance ceiling and quantify the limitations of pixel-only classification.

EfficientNetB0 with ImageNet weights was trained on 128×128 upscaled CIFAKE images using the two-stage protocol. The result — 67.48% accuracy, 0.8278 AUC-ROC, and specificity of only 0.4016 — confirmed that spatial features at this resolution are insufficient for reliable discrimination. The model predicted 15,465 of 20,000 test images as FAKE, revealing a severe false-positive bias. This result provided concrete quantitative motivation for frequency-domain analysis.

C. Experiment 2: Frequency-Only CNN

Objective: Determine whether DCT spectral features alone outperform spatial pixel features.

An EfficientNetB0 backbone was trained on on-the-fly DCT maps using an identical architecture and training protocol to the spatial baseline. The Dense(128) adapter head was added to enable frozen-phase convergence. Results: 83.26% accuracy, 0.9140 AUC-ROC, and specificity 0.8584. The +10.3% AUC improvement and +45.7% specificity improvement over the spatial baseline confirmed that DCT features carry discriminative information that spatial pixels do not, and nearly eliminated the false-positive problem.

D. Experiment 3: Dual-Stream EfficientNetB0 Fusion

Objective: Test whether fusing spatial and frequency features outperforms either single stream.

Two parallel EfficientNetB0 backbones (one RGB, one DCT) were trained with the late-fusion architecture. Result: 85.41% accuracy, 0.9305 AUC-ROC. Fusion outperformed both the spatial-only (0.8278 AUC) and frequency-only (0.9140 AUC) configurations, confirming the complementary nature of the two information streams and validating the core hypothesis of the dual-stream approach.

E. Experiment 4: Xception Backbone Upgrade

Objective: Test whether upgrading from EfficientNetB0 to Xception, with larger input resolution, produces a substantial accuracy gain.

Both backbones were replaced with Xception at 299×299 input. Xception's Depthwise Separable Convolutions provide significantly greater capacity for capturing point-like micro-texture anomalies. The result was transformative: 96.37% accuracy, 0.9944 AUC-ROC, 0.9580 specificity — a near-perfect classifier on CIFAKE. This 11-point accuracy improvement confirmed that backbone quality dominates fusion architecture at this task scale.

F. Experiment 5: FaceForensics++ Domain Adaptation

Objective: Extend detection capability to GAN-based face manipulation without forgetting CIFAKE knowledge.

The CIFAKE-trained Xception model was fine-tuned on FF++ C23 frames using extremely conservative hyperparameters (lr= 10^{-6}, top-10 layers unfrozen, 10 epochs). Results: 86.0% accuracy, 0.9240 F1-Score, 0.7389 AUC-ROC, 0.9897 recall. The high recall (98.97%) is appropriate for forensic deployment — the model misses very few fake images — at the cost of lower specificity (0.0670).

G. Experiment 6: Zero-Shot Generalization

Objective: Stress-test the model against 25 generators not seen during training to quantify the limits of cross-generator generalization.

The FF++-adapted model was evaluated on ArtiFact+COCO2017 without retraining. Results: 57.62% accuracy, 0.5557 AUC-ROC. The model correctly identified 88.54% of fake images (high recall) but incorrectly flagged 73.24% of real images (low specificity), indicating that the model learned generator-specific spectral fingerprints rather than universal AI generation mathematics. This result is an honest quantification of the system's current limitations and directly motivates multi-generator training as future work.

H. Ablation Study: Stream Contribution

The progressive improvement across experiments constitutes a structured ablation of each system component's contribution:

- *Spatial stream only*: AUC 0.8278
- *Frequency stream only*: AUC 0.9140 **(+10.3%)**
- *Dual-stream fusion (EfficientNetB0)*: AUC 0.9305 **(+1.65% over freq. only)**
- *Dual-stream fusion (Xception)*: AUC 0.9944 **(+6.39% over EffNet dual)**

This ablation confirms that both the frequency stream and the backbone upgrade contribute independently and additively to final performance.

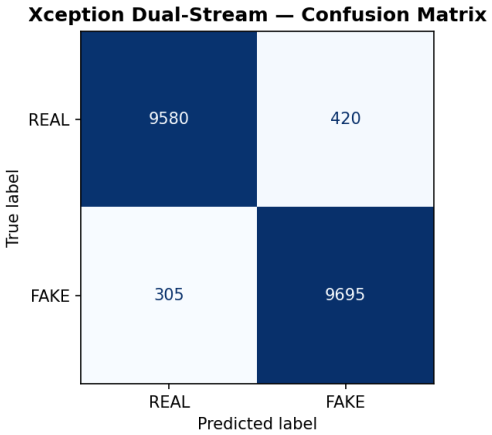


Fig. 14: Confusion matrix for the Dual-Stream Xception model on the CIFAKE 20,000-image test set. True positives (correct FAKE detections) and true negatives (correct REAL identifications) dominate, with very few misclassifications in either direction.

VI. Results and Discussion

A. Master Performance Comparison

Table IV presents the complete performance metrics across all six experimental configurations. The progression from Spatial CNN to Dual-Stream Xception represents the core scientific narrative: each architectural decision is empirically motivated by the previous result.

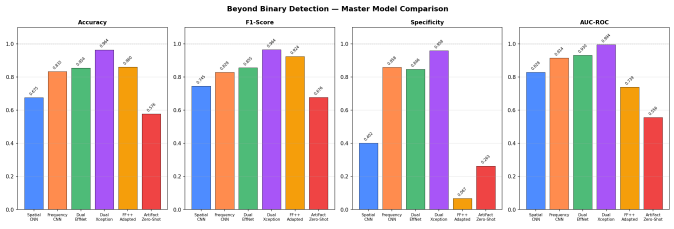


Fig. 12: Comprehensive Comparison of All Six Experiments Across Six Evaluation Metrics.

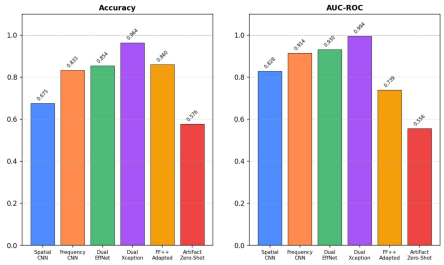


Fig. 13: AUC-ROC and accuracy comparison across all six experimental configurations, showing progressive improvement from the Spatial CNN baseline to the Dual-Stream Xception model.

B. Confusion Matrix Analysis

Key observations from Table IV:

Frequency CNN vs. Spatial CNN: The single most impactful improvement in the research was switching the input representation from RGB pixels to DCT maps, with an

TABLE IV: Master Performance Comparison Across All Experimental Configurations

Metric	Spatial CNN	Frequency CNN	Dual EffNet	Dual Xception	FF++ Adapted	ArtiFact (0-shot)
Accuracy	0.6748	0.8326	0.8541	**0.9637**	0.8600	0.5762
Precision	0.6131	0.8507	0.8483	**0.9585**	0.8664	0.5468
Recall	0.9481	0.8068	0.8625	0.9695	**0.9897**	0.8854
F1-Score	0.7446	0.8282	0.8554	**0.9640**	0.9240	0.6761
Specificity	0.4016	0.8584	0.8458	**0.9580**	0.0670	0.2676
AUC-ROC	0.8278	0.9140	0.9305	**0.9944**	0.7389	0.5557

identical backbone and training protocol. Specificity improved from 0.4016 to 0.8584 (+45.7 points), nearly eliminating the false-positive problem, while AUC improved by +10.3%.

Dual-Stream vs. Single-Stream: Late fusion of both streams consistently outperformed either stream in isolation. At EfficientNetB0 scale, fusion added +1.65% AUC over the frequency-only stream. This confirms complementary information content: spatial and spectral features are not redundant.

Xception vs. EfficientNetB0: The backbone upgrade was the single largest architectural contributor, adding +11.0 accuracy points and +6.39% AUC on the same dual-stream fusion task. Xception's depthwise separable convolutions and larger 299×299 input resolution provide substantially greater capacity for micro-texture discrimination.

Domain Adaptation: The 10.4-point accuracy drop and 25.5-point AUC drop from CIFAKE to FF++ reflects the substantial domain shift between diffusion-based and GAN-based fakes. Nevertheless, 86.0% accuracy and 0.9897 recall on a completely different manipulation domain demonstrates meaningful cross-architecture transfer.

Zero-Shot Generalization: AUC 0.5557 against 25 unseen generators confirms that the model learned generator-specific spectral fingerprints. Critically, the high recall (0.8854) indicates the model is conservative in the forensically correct direction — it tends to flag images as fake rather than miss fakes — but at the cost of excessive false positives on real images (specificity 0.2676).

C. Grad-CAM Qualitative Analysis

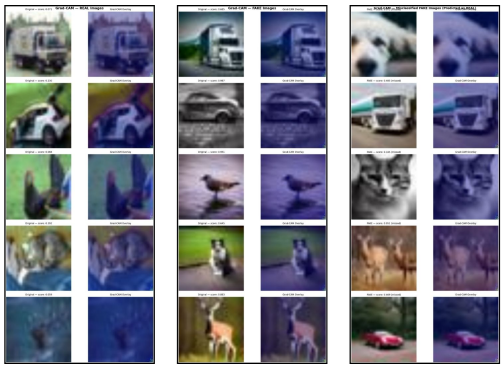


Fig. 15: Grad-CAM heatmap visualizations. Left: correctly classified real image (low activation, diffuse heatmap). Center: correctly classified fake image (high activation localizing synthetic artifacts). Right: misclassified fake image (false negative), revealing regions where DCT fingerprints were too subtle for detection.

Grad-CAM heatmaps reveal qualitatively meaningful patterns. For correctly classified fake images, activations concentrate on regions with visible or spectral inconsistencies (boundary regions, uniform background areas, eye reflections). For correctly classified real images, activations are diffuse and low-magnitude, reflecting the absence of structured artifact signals. Most forensically valuable, heatmaps for misclassified fakes (false negatives) reveal the model's failure modes: synthetic images where the spectral fingerprints are subtle and spatially distributed, preventing any single region from exceeding the detection threshold.

D. Error Analysis

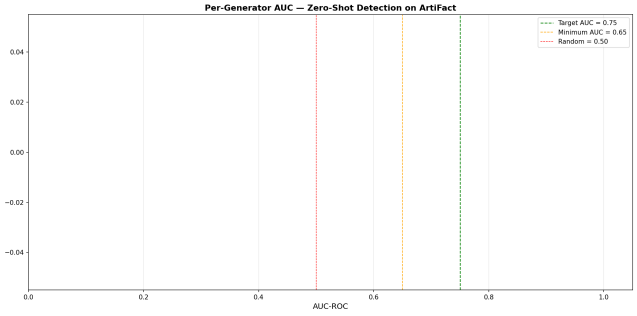


Fig. 16: Per-generator AUC-ROC breakdown for the zero-shot ArtiFact evaluation. Generator families show varying detectability: older GAN architectures (ProGAN, StyleGAN1) yield higher AUC, while modern diffusion models and StyleGAN3 are hardest to detect without targeted training.

The zero-shot evaluation revealed a systematic pattern: the model performs better against generator families sharing architectural similarities with its training data. Older GAN architectures (ProGAN, BigGAN) that rely on transposed convolution upsampling leave stronger periodic artifacts in the frequency domain, leading to higher per-generator AUC values. Modern diffusion models (DDPM, Latent Diffusion, VQ-Diffusion) and StyleGAN3 — which employ alias-free upsampling specifically designed to suppress frequency-domain fingerprints — are hardest to detect. This analysis directly motivates multi-generator training as the primary future direction.

VII. Limitations and Future Work

A. Current Limitations

Generator-specific overfitting: The model learned spectral fingerprints specific to CIFAKE (Stable Diffusion v1.4) and six FF++ GAN types. Zero-shot AUC of 0.5557 against 25 unseen generators confirms insufficient generalization to the broader AI generation landscape.

CIFAKE resolution: CIFAKE's native 32×32 resolution is atypical of real-world forensic targets (typically high-resolution photographs), potentially inflating performance metrics relative to real deployment scenarios.

Imbalanced face detection in FF++: Haar-cascade face detection produced an imbalanced extraction (8,090 real vs. 4,832 fake training frames) due to detection failures in certain manipulation types, potentially biasing the domain-adapted model toward over-predicting fakes.

Single-frame analysis: The system classifies individual images rather than video sequences, missing temporal consistency cues available in video-level deepfake detection.

B. Future Work

Multi-generator training: Incorporating ArtiFact's 25 generators into the training set alongside CIFAKE and FF++ is the highest-priority extension. Curriculum learning (starting with generators most similar to existing training data) may accelerate convergence.

Frequency-specialized backbone: Replacing the second Xception stream with a frequency-selective architecture (e.g., a shallow MLP or spectral CNN) may better exploit the non-spatial structure of DCT coefficient maps.

Higher-resolution evaluation: Testing on DALL-E 3, Midjourney v6, and Stable Diffusion XL outputs at native 1024×1024 resolution is critical for assessing real-world robustness against state-of-the-art synthesis.

Attention-based fusion: Replacing late concatenation with a cross-attention module would allow the model to dynamically weight the contribution of each stream based on image content, potentially improving performance on generators with varying artifact distributions.

Adversarial robustness: Testing against adversarially perturbed inputs designed to suppress DCT fingerprints is critical for deployment in adversarial forensic contexts.

VIII. Conclusion

This paper addressed the dual challenges of spectral blindness and interpretability opacity in deepfake detection. We proposed a dual-stream frequency-aware CNN that simultaneously processes spatial RGB features and DCT spectral representations through parallel Xception backbones, fusing both streams via late concatenation before binary classification.

The key finding is unambiguous: frequency-domain analysis is superior to spatial-only analysis for detecting AI-generated images, achieving AUC-ROC 0.9140 versus 0.8278 for a spatial-only baseline — driven entirely by changing the input representation from pixels to DCT coefficients with an otherwise identical model. Combining both streams via late fusion consistently outperformed either stream alone, confirming that spatial and spectral features carry complementary discriminative information. The Xception backbone, with its depthwise separable convolutions and 299×299 input, achieved near-perfect CIFAKE performance at 96.37% accuracy and 0.9944 AUC-ROC.

Grad-CAM explainability, implemented with a novel Keras 3 compatibility fix, successfully transforms binary predictions into spatial forensic evidence maps, revealing which image regions drove each classification decision. Domain adaptation to FaceForensics++ C23 demonstrated that the model's learned spectral representations transfer to GAN-based face manipulation, achieving 86.0% accuracy with 0.9897 recall. Honest zero-shot evaluation against 25 unseen generators quantified the model's current generalization limits (AUC 0.5557), directing future work toward multi-generator training.

Finally, the Stream-and-Clean decoupled ingestion pipeline demonstrated that forensic-grade deepfake research on 100+ GB datasets is feasible within a 15 GB cloud storage constraint on free-tier hardware, providing a reproducible blueprint for resource-constrained AI forensics research.

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